Harry Nguyen

Vancouver, British Columbia

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Technical Skills

Software Tools: Altium, LTspice, MATLAB, Simulink, Visual Studio, SolidWorks, Excel, PowerPoint
Programming: C/C++, Java, Python, ARM Assembly, Verilog
Hardware: DC/BLDC motors, Microcontrollers, Signal Generator, Oscilloscope, Multimeter, SMT/THT Soldering

Education

University of British Columbia May 2027 (Expected)
Bachelor of Applied Science, Electrical Engineering *Vancouver, BC*

- **COOP:** Available for 12 months beginning May 2026
- **Relevant Courses:** Engineering Electromagnetic Fields and Waves, Electro-Mechanical Energy Conversion and Transmission, Circuit Analysis, Electronic Circuit Design, Data Structures and Algorithms

Engineering Design Team

Open Robotics, UBC Vancouver Sep. 2025 – Present
Engineering Design Team Member *Vancouver, BC*

- Co-developed **HapticBot**, an educational haptic knob simulating electro-mechanical effects of RLC components and diodes
- Conducted **system-level testing and debugging** of prototype hardware, characterizing BLDC motor efficiency, torque, and encoder feedback performance
- Designed a custom **4-layer PCB in Altium** with integrated current sensor, encoder, motor driver, and ESP32 MCU

Technical Projects

Piano-Playing Robot Jan. 2026 – Apr. 2026
Skills: *LTspice, Altium Designer, PCB Layout, Power Electronics, Gate Drivers*

- Designed and routed mixed-signal **4-layer PCBs in Altium** using a **mother-daughterboard architecture**, applying strategic power stackup to optimize high-current delivery
- Simulated **IR2184 half-bridge gate driver** circuits in **LTspice** to verify MOSFET transient response and dead-time, resolving failures prior to physical prototyping
- Applied **DFM practices** integrating M3 clearances, inline test points, and JLCPCB rules to generate finalized **BOM, Gerber and NC Drill fabrication packages**

Wisdom Dispenser / Photo Frame — Personal Engineering Project May 2026
Skills: *ESP32-S3, Arduino, C++, SPI, Embedded Systems, Protoboard Assembly*

- Designed and built a custom **embedded device** on a hand-soldered protoboard featuring a 3.5” ILI9488 SPI TFT display, SD card photo frame mode, 3-button UI, and USB-C power circuit
- Troubleshot **SPI bus conflicts** and library incompatibilities between Arduino GFX and ESP32 core versions to achieve stable concurrent display and SD card operation on shared GPIO
- Architected **C++ firmware** with persistent flash storage, multi-mode UI state machine, JPEG decoding, and progressive content unlock system

Autonomous Coin-Picking Robot Mar. 2025 – Apr. 2025
Skills: *C Programming, ATMEGA328P, Analog Circuit Design, PWM Control*

- Integrated a 2-microcontroller system (**ATMEGA328P** and **EFM8**) to control robot operation via JDY-40 radio module for autonomous/remote operation within a set boundary
- Designed and troubleshot **analog sensor circuits** (**Colpitts Oscillator**, **CMOS inverters**) to detect coins and boundaries with **100% success rate**
- Implemented **C** control logic with joystick inputs, radio comms, and **PWM actuators** for efficient autonomous behavior

Microcontroller-Controlled Reflow Oven Feb. 2025 – Mar. 2025
Skills: *Assembly, Finite State Machines, Embedded Systems, N76E003 MCU*

- Drove a reflow oven using **N76E003 MCU** and **OP07 op-amp**, achieving **$\pm 3^\circ$C accuracy** over a 25–240°C heating profile
- Applied **Assembly FSM logic** to manage 5-stage soldering profiles with automated, customizable soldering cycles
- Developed intuitive UI with **LCD** and pushbuttons for dynamic profile adjustment and real-time temperature monitoring

Other Work Experience

DECO Windshield Repair Apr. 2024 – Aug. 2025
Site Operator *Vancouver, BC*

- Executed precision structural repairs on laminated glass (chips and >24” cracks) utilizing low-speed drilling and vacuum-pressure resin injection cycles
- Applied meticulous attention to detail to mitigate fracture propagation using stress-relief drilling to contain damage within failure margins
- Achieved and maintained a repair plan sale rate of **>80%** and high customer satisfaction